

Chapter 14 - Hibernate Query Language (HQL)

1. What is HQL?

Hibernate Query Language (HQL) is an object-oriented query language. It works with entity names and entity properties rather than database table and column names.

SQL	HQL
Uses table names	Uses entity names
Uses column names	Uses entity properties
Database specific	Database independent
Returns rows	Returns entity objects or scalar values

2. Query Execution Flow

```
Application
|
createQuery(HQL)
|
Hibernate parses HQL
|
Generates SQL
|
Database
|
Entity Objects Returned
```

3. Basic Queries

```
from Employee

from Employee where salary > 50000

select e.name from Employee e

from Employee e order by e.name

select count(e) from Employee e
```

4. Named Parameters

Named parameters improve readability and help avoid SQL injection.

```
Query<Employee> q =
session.createQuery(
"from Employee where dept=:dept",
Employee.class);

q.setParameter("dept", "IT");
```

5. Aggregate Functions

HQL supports count(), sum(), avg(), min() and max().

6. Joins

HQL supports inner join, left join and fetch join. A fetch join can initialize associations in the same query to reduce additional selects.

```
from Department d
join fetch d.employees
```

7. Pagination

Pagination is implemented using `setFirstResult()` and `setMaxResults()`.

```
query.setFirstResult(0);
query.setMaxResults(10);
```

8. Update & Delete HQL

```
update Employee
set salary=salary+1000
where dept='IT';

delete from Employee
where status='INACTIVE';
```

9. SQL Generation

Hibernate translates HQL into database-specific SQL before execution. You write HQL once and Hibernate generates the appropriate SQL for the configured database.

10. Best Practices

- Use named parameters.
- Use pagination for large result sets.
- Use fetch joins carefully to avoid unnecessary data loading.
- Prefer typed queries where available.

11. Interview Questions

1. Difference between HQL and SQL? 2. Does HQL use table names or entity names? 3. What is a fetch join? 4. How do you implement pagination? 5. How are parameters passed safely in HQL?